

Series1 (Solution)

1- The principal components of a computer are:

- Central Processing Unit (CPU):
- Principal Memories
- Motherboard:
- Input/ Output Devices:
- Bus

2. Explain the difference between volatile and non-volatile memory.

- **Volatile memory** (e.g., RAM) requires constant power to retain data. If the computer is turned off, the data is lost.
- **Non-volatile memory** (e.g., hard drives or flash memory) retains data even when the computer is turned off.

3. What are the main roles of the central processing unit (CPU) in a computer?

The CPU (processor) executes instructions from programs, performs calculations, processes data, and coordinates the activities of other hardware components in the computer.

4. What is a data bus and how does it work?

A data bus is a set of wires or parallel pathways through which data is transmitted between components of a computer, such as the processor, memory, and peripherals. It enables data transfer and communication between these parts.

5. Describe the main functions of the following components: processor, RAM, ROM, and hard disk.

- **Processor (CPU):** Executes instructions and processes data.
- **Random Access Memory (RAM):** Temporarily stores data that the processor uses actively, enabling quick access.
- **ROM (Read-Only Memory)** It is crucial for the boot process and basic system functions. It is a non-volatile memory that:
 - **Stores permanent data** like firmware.
 - **Retains data when powered off.**
 - **Holds read-only data**, typically written during manufacturing.
- **Hard disk:** Provides long-term storage for data, including the operating system, applications, and user files.

6. What is the difference between an input device and an output device? Give examples.

- An **input device** allows users to send data to the computer (e.g., keyboard, mouse).
- An **output device** displays or returns processed information from the computer to the user (e.g., monitor, printer).

7. What is a motherboard and what are its main functions?

The motherboard is the main printed circuit board of a computer. It connects and allows communication between the processor, memory, peripherals, and other components, while also housing ports for connecting external devices.

8. What are the characteristics of Von Neumann architecture?

The Von Neumann architecture is characterized by a single memory that stores both data and instructions, a central processing unit (CPU) to execute these instructions, and input/output mechanisms to interact with external devices.

9. What is an instruction cycle and what are its steps?

An instruction cycle is the process by which the processor fetches, decodes, and executes an instruction. The steps are:

1. **Fetch:** The processor retrieves the instruction from memory.
2. **Decode:** The processor interprets the instruction.
3. **Execute:** The processor performs the operation specified by the instruction.

10. **What is a binary system and why is it used in computers?**
The binary system is a numbering system that uses only two digits: 0 and 1. It is used in computers because they operate with electronic circuits that have two states: on (represented by 1) and off (represented by 0). This makes binary very efficient for data processing and storage in digital systems.
11. **How many states can be represented with 3 bits? with 4 bits? and with n bits in general?**
- With 3 bits, we can represent $2^3 = 8$ different states.
 - With 4 bits, we can represent $2^4 = 16$ different states.
 - And with n bits in general, we can represent 2^n different states.
12. **How to write a number a in base a?**
 $(a)_{10} = (10)_a$ ex : $(2)_{10} = (10)_2$, $(8)_{10} = (10)_8 \dots$
13. **Using three bits, show the binary counting sequence from 000 to 111.**

Decimal	Binary (3 bits)
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

14. **What is the maximum number that we can count up to using 10 bits?**

The maximum number that can be represented using 10 bits is calculated using the formula:
$$\text{Maximum value} = 2^n - 1$$
where n is the number of bits.
For 10 bits:
$$2^{10} - 1 = 1024 - 1 = 1023$$

15. **How many bits are needed to count up to a maximum of 63?**

We need to find n such that:
$$2^n - 1 \geq 63$$
Steps to Find n :

1. Rearranging the inequality gives:
$$2^n \geq 64$$
2. Since $64 = 2^6$, it follows that:
$$n \geq 6$$

16. **What is the largest decimal value that can be represented by**

(a) Eight-bit binary number:
For $n = 8$:
$$2^8 - 1 = 256 - 1 = 255$$

(b) Sixteen-bit binary number:
For $n = 16$:
$$2^{16} - 1 = 65536 - 1 = 65535$$

Exercise1:

1- Here is a data sheet with the correct units of measurement:

- **Intel Core™ i5** (frequency 3.40 **GHz**, cache memory 4 **MB**)
- **Windows 8.1** 64 **bit**
- **RAM 4 GB** with frequency of 1333 **MHz**
- **Hard disk 850 GB**, transfer rate 4 **Gbps**
- **Integrated network card (LAN): 100 Mbps**
- **ADSL connection 2 Mbps**
- **WebCam: resolution 12 MP (megapixels)**

2- Perform the conversions to the appropriate units for the following values:

1. 300 MHz = ? GHz 1 GHz = 1000 MHz 300 MHz = 0.3 GHz 2. 2.89 GHz = ? KHz 1 GHz = 1,000,000 KHz 2.89 GHz = 2,890,000 KHz 3. 1.8 GHz = ? Hz 1 GHz = 1,000,000,000 Hz 1.8 GHz = 1,800,000,000 Hz 4. 500 Go = ? To 1 To = 1024 Go 500 Go = 0.488 To 5. 3072 Mo = ? Go 1 Go = 1024 Mo 3072 Mo = 3 Go	6. 50 o = ? Bit 1 o = 8 Bits 50 o = 50 × 8 = 400 Bits 7. 4 Ko = ? Bit 1 Ko = 1024 o = 1024 × 8 = 8192 Bits 4 Ko = 4 × 8192 = 32,768 Bits 8. 2³⁰ Bits = ? Mo 1 Mo = 8,388,608 Bits 2³⁰ Bits = 1 Go = 1024 Mo 9. 1 To = ? Ko 1 To = 1,024 Go 1 Go = 1,024 Mo 1 Mo = 1,024 Ko 1 To = 1,024 × 1,024 × 1,024 = 1,048,576 Ko
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Exercise2: A student wants to transfer software to a storage unit such as:

1- USB Drive Capacity > Software Size (8 GB) > (4 GB) Therefore, the student can store the software on the USB drive. 2- The capacity of a CD is 1400 MB Software Size: 4 GB = 4 × 1024 MB Number of CDs = (4 × 1024 MB) / (1400 MB) ≈ 2,93 So, it would take 3 CDs. 3- Conclusion : Comparing the two solutions, we can say that it is more convenient for the student to use 1 USB drive to store the software than to use 3 CDs..
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Exercise3: Express each of the given numbers in polynomial form based on their respective bases:

$$(101101)_2 = 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0$$

$$(1257)_{10} = 1 \cdot 10^3 + 2 \cdot 10^2 + 5 \cdot 10^1 + 7 \cdot 10^0$$

$$(76)_8 = 7 \cdot 8^1 + 6 \cdot 8^0$$

$$(F15B)_{16} = 15 \cdot 16^3 + 1 \cdot 16^2 + 5 \cdot 16^1 + 11 \cdot 16^0$$

Exercise4:

1- Find the bases x and y such that: $(403)_x = (103)_{10}$, $(147)_y = (103)_{10}$

$$(403)_x = (103)_{10}, \quad 3 \times x^0 + 0 \times x^1 + 4 \times x^2 = 103, \quad 4 \times x^2 = 100 \Rightarrow x = \pm 5 \quad \text{So } x=5 \text{ because a base cannot be negative.}$$

$$(147)_y = (103)_{10}, \quad 7 \times y^0 + 4 \times y^1 + 1 \times y^2 = 103, \quad -96 + 4y + y^2 = 0 \Rightarrow \Delta = 400, \quad y_1 = -12 \text{ et } y_2 = +8$$

So $y=8$ because a base cannot be negative.

2- Find the base b

1. $(1001)_b = (201)_{16}$

- $(201)_{16}$ is a number in base 16.
- Let's convert $(201)_{16}$ to base 10 to get a decimal value:

$$201_{16} = 2 \times 16^2 + 0 \times 16^1 + 1 \times 16^0$$

$$201_{16} = 2 \times 256 + 0 \times 16 + 1 = 512 + 0 + 1 = 513_{10}$$

- Now we know that $(1001)_b = 513_{10}$.

In base b, 1001_b means:

$$1 \times b^3 + 0 \times b^2 + 0 \times b^1 + 1 \times b^0 = 513$$

This gives the equation:

$$b^3 + 1 = 513$$

$$b^3 = 512$$

$$b = \sqrt[3]{512} = 8$$

Answer: $b = 8$

2. $(1001100)_b = (114)_8$

- $(114)_8$ is a number in base 8.
- Let's convert $(114)_8$ to base 10:

$$114_8 = 1 \times 8^2 + 1 \times 8^1 + 4 \times 8^0$$

$$114_8 = 1 \times 64 + 1 \times 8 + 4 = 64 + 8 + 4 = 76_{10}$$

- Now we know that $(1001100)_b = 76_{10}$.

In base b, 1001100_b means:

$$1 \times b^6 + 0 \times b^5 + 0 \times b^4 + 1 \times b^3 + 1 \times b^2 + 0 \times b^1 + 0 \times b^0 = 76$$

This gives the equation:

$$b^6 + b^3 + b^2 = 76$$

This is a polynomial equation in b, and after some trials, we find that $b = 3$ satisfies the equation.

Answer: $b = 3$

3. $(125)_b = (293)_{10}$

- Let's convert $(125)_b$ to base 10:

In base b, 125_b means:

$$1 \times b^2 + 2 \times b^1 + 5 \times b^0 = 293$$

This gives the equation:

$$b^2 + 2b + 5 = 293$$

Let's solve the quadratic equation:

$$b^2 + 2b - 288 = 0$$

Using the quadratic formula:

$$b = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-288)}}{2(1)}$$

$$b = \frac{-2 \pm \sqrt{4 + 1152}}{2}$$

$$b = \frac{-2 \pm \sqrt{1156}}{2}$$

$$b = \frac{-2 \pm 34}{2}$$

The solutions are:

$$b = \frac{-2 + 34}{2} = 16$$

$$b = \frac{-2 - 34}{2} = -18 \text{ (not a valid solution for a base)}$$

Therefore, $b = 16$.

Answer: $b = 16$